Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Bullock's Oriole"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                      paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjus
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

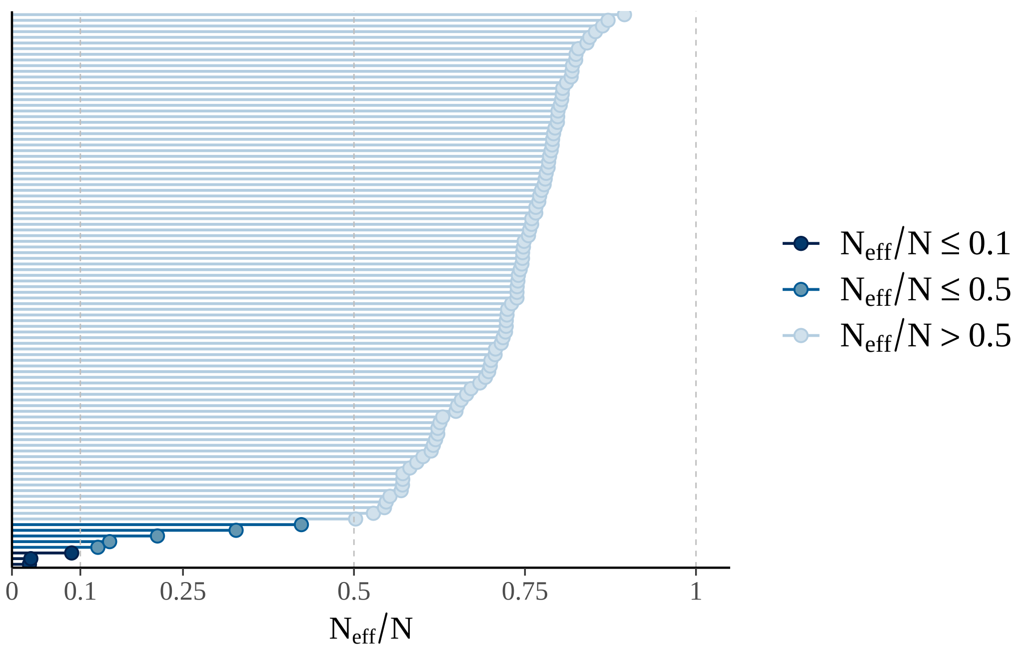
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_eff to N (number of samples), the better.




```
[1] "effective sample size too small"
```

```
[1] "n_eff ratio bulk ESS:"
```

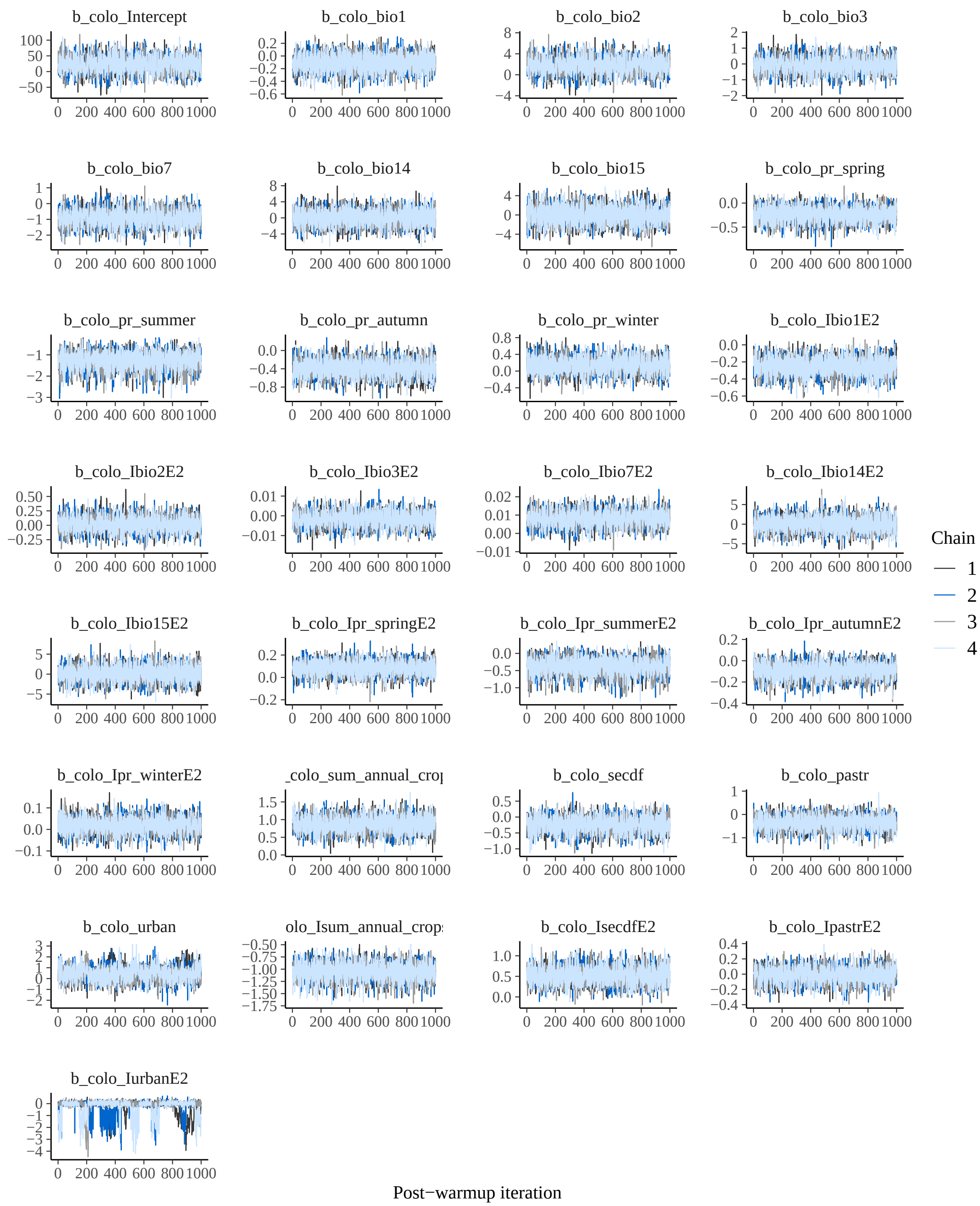
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.0256  0.8567  1.0401  1.0184  1.1937  1.8182 
```

```
[1] "n_eff ratio tail ESS:"

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.07542 0.66956 0.73998 0.71052 0.78858 0.89543

[1] "effective sample size in bulk or tail too small"
```

Do chains mix well:

```
No divergences to plot.
No divergences to plot.
```


```
No divergences to plot.
```

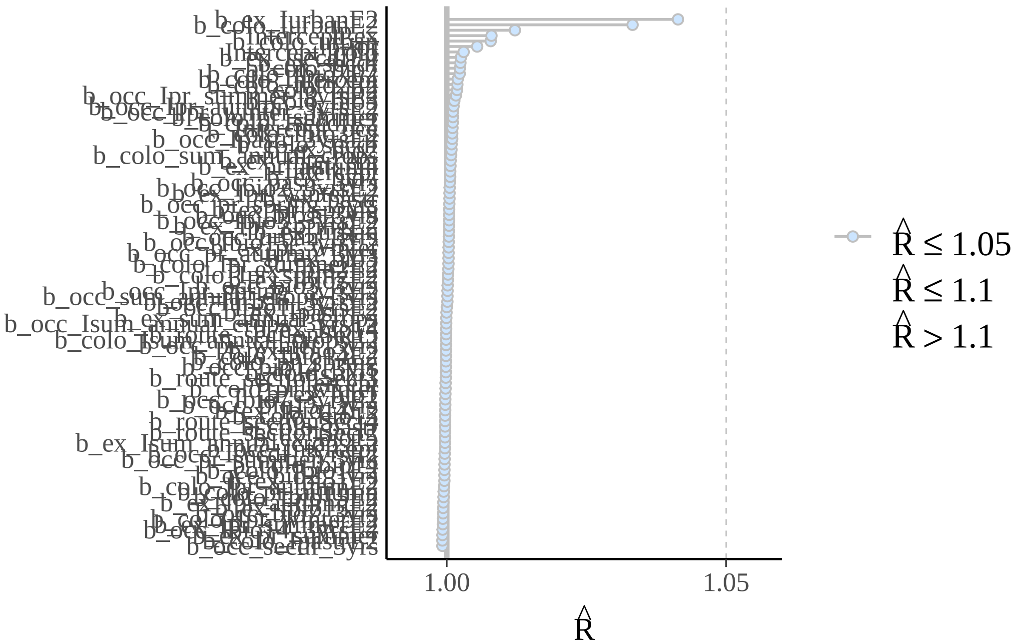


b_colo_Intercept
b_colo_bio1
b_colo_bio2
b_colo_bio3
b_colo_bio7
b_colo_bio14
b_colo_bio15
b_colo_pr_spring
b_colo_pr_summer
b_colo_pr_autumn
b_colo_pr_winter
b_colo_Ibio1E2
b_colo_Ibio2E2
b_colo_Ibio3E2
b_colo_Ibio7E2
b_colo_Ibio14E2
b_colo_Ibio15E2
b_colo_Ipr_springE2
b_colo_Ipr_summerE2
b_colo_Ipr_autumnE2
b_colo_Ipr_winterE2
_colo_sum_annual_cro
b_colo_secdf
b_colo_pastr
b_colo_urban
olo_Isum_annual_crop
b_colo_IsecdfE2
b_colo_IpastrE2
b_colo_IurbanE2
Chain
1
2
3
4
Post−warmup iteration


b_ex_Intercept
b_ex_bio1
b_ex_bio2
b_ex_bio3
b_ex_bio7
b_ex_bio14
b_ex_bio15
b_ex_pr_spring
b_ex_pr_summer
b_ex_pr_autumn
b_ex_pr_winter
b_ex_Ibio1E2
b_ex_Ibio2E2
b_ex_Ibio3E2
b_ex_Ibio7E2
b_ex_Ibio14E2
b_ex_Ibio15E2
b_ex_Ipr_springE2
b_ex_Ipr_summerE2
b_ex_Ipr_autumnE2
b_ex_Ipr_winterE2
b_ex_sum_annual_crops
b_ex_secdf
b_ex_pastr
b_ex_urban
ex_Isum_annual_cropsI
b_ex_IsecdfE2
b_ex_IpastrE2
b_ex_IurbanE2
Chain
1
2
3
4
Post−warmup iteration

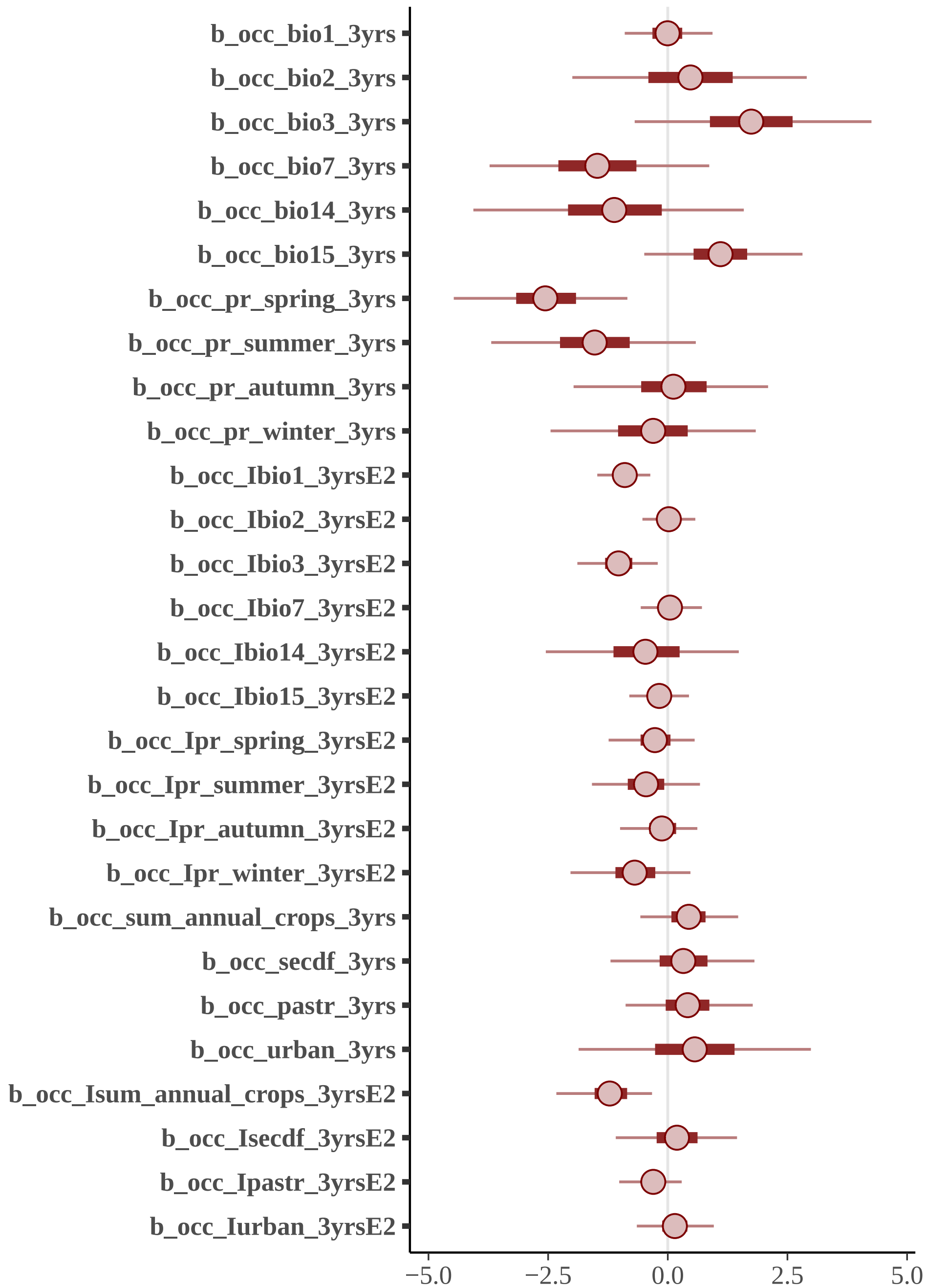
Are R-hat values fine:

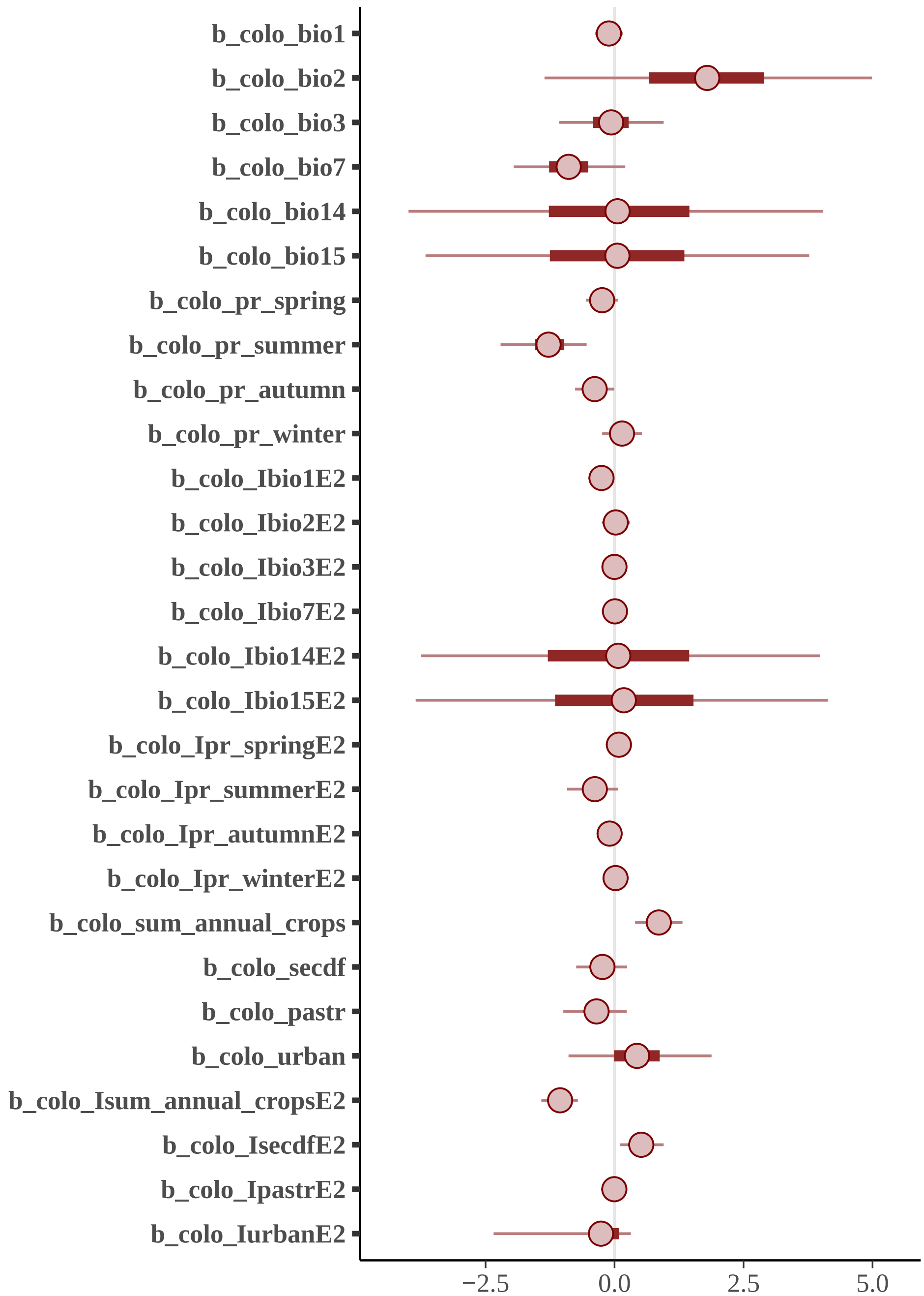
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.9992  0.9997  1.0002  1.0014  1.0008  1.0414 
```

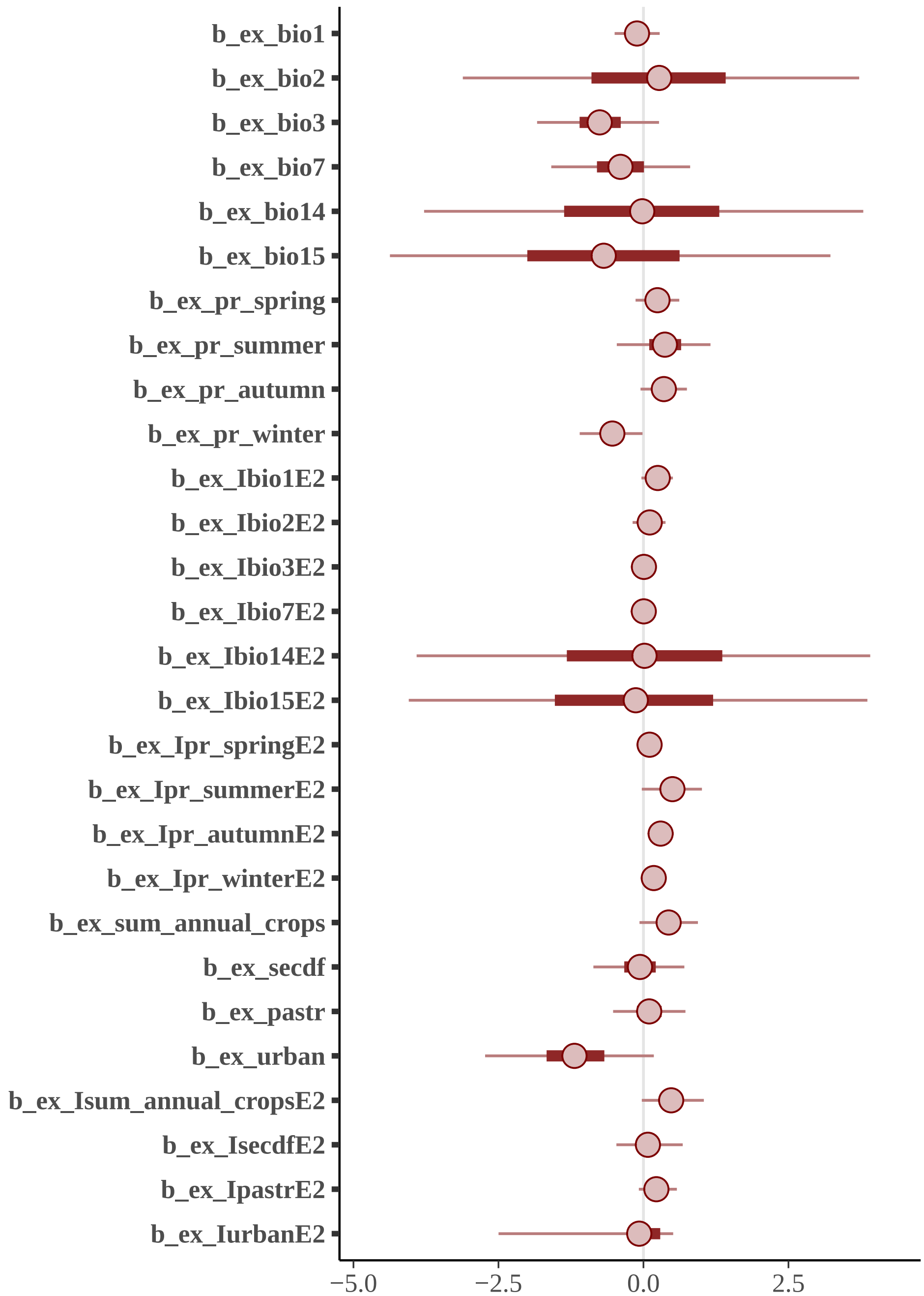


Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"

[1] "b_occ_pr_spring_3yrs"           "b_occ_Ibio1_3yrsE2"            
[3] "b_occ_Ibio3_3yrsE2"             "b_occ_Isum_annual_crops_3yrsE2"

[1] "b_colo_pr_summer"           "b_colo_pr_autumn"          
[3] "b_colo_Ibio1E2"             "b_colo_sum_annual_crops"   
[5] "b_colo_Isum_annual_cropsE2" "b_colo_IsecdfE2"           

[1] "b_ex_pr_winter"    "b_ex_Ibio3E2"      "b_ex_Ipr_autumnE2"
[4] "b_ex_Ipr_winterE2"
```


b_occ_bio1_3yrs
b_occ_bio2_3yrs
b_occ_bio3_3yrs
b_occ_bio7_3yrs
b_occ_bio14_3yrs
b_occ_bio15_3yrs
b_occ_pr_spring_3yrs
b_occ_pr_summer_3yrs
b_occ_pr_autumn_3yrs
b_occ_pr_winter_3yrs
b_occ_Ibio1_3yrsE2
b_occ_Ibio2_3yrsE2
b_occ_Ibio3_3yrsE2
b_occ_Ibio7_3yrsE2
b_occ_Ibio14_3yrsE2
b_occ_Ibio15_3yrsE2
b_occ_Ipr_spring_3yrsE2
b_occ_Ipr_summer_3yrsE2
b_occ_Ipr_autumn_3yrsE2
b_occ_Ipr_winter_3yrsE2
b_occ_sum_annual_crops_3yrs
b_occ_secdf_3yrs
b_occ_pastr_3yrs
b_occ_urban_3yrs
b_occ_Isum_annual_crops_3yrsE2
b_occ_Isecdf_3yrsE2
b_occ_Ipastr_3yrsE2
b_occ_Iurban_3yrsE2
−5.0
−2.5
0.0
2.5
5.0


b_colo_bio1
b_colo_bio2
b_colo_bio3
b_colo_bio7
b_colo_bio14
b_colo_bio15
b_colo_pr_spring
b_colo_pr_summer
b_colo_pr_autumn
b_colo_pr_winter
b_colo_Ibio1E2
b_colo_Ibio2E2
b_colo_Ibio3E2
b_colo_Ibio7E2
b_colo_Ibio14E2
b_colo_Ibio15E2
b_colo_Ipr_springE2
b_colo_Ipr_summerE2
b_colo_Ipr_autumnE2
b_colo_Ipr_winterE2
b_colo_sum_annual_crops
b_colo_secdf
b_colo_pastr
b_colo_urban
b_colo_Isum_annual_cropsE2
b_colo_IsecdfE2
b_colo_IpastrE2
b_colo_IurbanE2
−2.5
0.0
2.5
5.0


b_ex_bio1
b_ex_bio2
b_ex_bio3
b_ex_bio7
b_ex_bio14
b_ex_bio15
b_ex_pr_spring
b_ex_pr_summer
b_ex_pr_autumn
b_ex_pr_winter
b_ex_Ibio1E2
b_ex_Ibio2E2
b_ex_Ibio3E2
b_ex_Ibio7E2
b_ex_Ibio14E2
b_ex_Ibio15E2
b_ex_Ipr_springE2
b_ex_Ipr_summerE2
b_ex_Ipr_autumnE2
b_ex_Ipr_winterE2
b_ex_sum_annual_crops
b_ex_secdf
b_ex_pastr
b_ex_urban
b_ex_Isum_annual_cropsE2
b_ex_IsecdfE2
b_ex_IpastrE2
b_ex_IurbanE2
−5.0
−2.5
0.0
2.5